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Interstitial Lung Disease and Sarcoidosis

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Key Points:

- Health disparities exist among socioeconomic status, race, gender, and geographic location in interstitial lung disease (ILD) and sarcoidosis.
- In ILD and sarcoidosis, access to care, stress, and environmental exposures are potential determinants of health disparities, but research is needed to identify additional modifiable determinants.
- Avenues to achieve health equity in ILD and sarcoidosis include health policy, research, and expanding collaborative clinical care models.

Abstract

Interstitial lung disease (ILD), a clinically recognized group of diseases resulting in pulmonary fibrosis, affects up to 200 individuals per 100,000 in the United States. Sarcoidosis has a wide range of clinical manifestations including pulmonary fibrosis. Health disparities are prevalent in both ILD and sarcoidosis around socioeconomic status, race, gender, and geographic location. This review outlines the known health disparities, discusses possible determinants of disparities, and outlines a path to achieve equity in ILD and sarcoidosis.

Interstitial lung disease (ILD) is a clinically recognized group of diseases that all result in scarring of the lung interstitium, also termed pulmonary fibrosis.[1, 2] Causes of ILD vary from known etiologies, such as connective-tissue disease (CTD-ILD) and hypersensitivity pneumonitis (HP), to unknown etiologies such as idiopathic pulmonary fibrosis (IPF) and sarcoidosis. When individually considered each ILD is a rare disease,[3] however, as a group the prevalence of ILD approaches 200 individuals per 100,000 in the United States (US) and has increased by 19.1% over the last decade, in step with an aging population.[4, 5] Sarcoidosis, while considered an ILD, has a range of pulmonary manifestations including airway obstruction[6, 7] with a prevalence between 126-160 per 100,000 individuals[8] and as high as 519 per 100,000 among Black females in the US.[9] Death and disability attributable to ILD is on the rise relative to other diseases,[4] suggesting that resource allocation may be inadequate.

Healthcare spending on ILD per patient is higher than the per patient cost of asthma and COPD and is increasing over time.[10] Diagnostic complexity of ILD and sarcoidosis leads to financial burden and patient stress.[11] Delays in diagnosis[11] and initial misdiagnoses add to unmeasured antecedent healthcare costs. Sarcoidosis often affects individuals among prime working years, resulting in missing work, loss of earnings[12] and disability[13] for some, which worsens the financial impact of the disease.

While there is variation in symptoms, quality of life, and disease course of ILD and sarcoidosis, one common thread is that health disparities are prevalent, and the burden of these diseases on patients and healthcare systems is significant. We review health disparities present in ILD and sarcoidosis by socioeconomic status, race, gender, and geographic location, discuss possible determinants of disparities, and outline a path to achieve equity.

Socioeconomic Disparities

The diagnosis of ILD or sarcoidosis associates with significant cost for all, and especially burdens low-income individuals.[14, 15] Careful and consistent pharmacotherapy is paramount to limiting fibrosis.[15, 16] However, treatment imposes a significant financial millstone that has increased over time.[11, 15, 17] Individuals with low income or individuals in low-income zip codes report barriers to obtaining medicines[18] and difficulty accessing medical care.[19] These barriers translate to poor outcomes. Individuals with IPF and low income have worse progression free survival[20] and experience death more often than lung transplant, regardless of disease severity or other transplant eligibility factors (Figure 1).[21] Individuals with sarcoidosis and low income have worse lung function,[18] radiographic stage,[22] health-related quality of life (HRQoL)[14] and higher healthcare utilization as compared to those with sarcoidosis and higher income.[8] In

addition to income, insurance differences associate with worse outcomes. Patients with IPF who are un-insured or publicly insured are less likely to receive lung transplantation or inpatient rehabilitation.[23] Patients with sarcoidosis who are un-insured or publicly insured have lower lung function,[18] more disease progression,[24] and worse dyspnea,[22] perhaps secondary to barriers to care.[18]

Racial Disparities

In the US, significant racial and ethnic disparities remain throughout healthcare[25] and prevail among patients with ILD and sarcoidosis. Black individuals with ILD and sarcoidosis have more advanced disease at time of diagnosis[26] and lower lung function[7, 27] compared to non-Hispanic White individuals. Black individuals with sarcoidosis have higher rates of hospitalization,[28] higher mortality,[29, 30] and more organ involvement[26] compared to non-Hispanic White individuals with sarcoidosis (Figure 2). Black individuals with sarcoidosis are prescribed higher doses of corticosteroids[31] compared to non-Hispanic White individuals, however, are less likely to report medication adherence.[32] Though less frequently diagnosed with IPF, Black individuals who receive this diagnosis die at younger ages relative to White patients with IPF.[33] Multiple determinants interact to produce racial disparities including but not limited to socioeconomics, institutional racism, differential access to medical care, increased stress, and differential environmental exposures.[34, 35]

Sex and Gender Disparities

Sex and gender modify chronic disease including ILD and sarcoidosis via biological, environmental and social/behavioral interactions.[36] There is a complex relationship between sex hormones and lung disease[37] and their role is understudied in ILD[38] and sarcoidosis.[39] Gender

interacts with environmental risk factors for the development of ILD and sarcoidosis, including but not limited to traditionally gendered occupations, military service, hobbies, or tasks that may lead to differential environmental exposures, and tobacco use.[40-42] Patients with IPF are approximately twice as likely to be male, and male sex is a risk factor in the IPF Gender-Age-Physiology calculation for predicting future lung function decline[43] and mortality risk at 1 year.[44] CTD-ILD conversely exhibits higher prevalence of females affected and higher female mortality.[45] Although more prevalent among females, CTD-ILD demonstrates a more progressive course in males, including higher frequency of usual interstitial pneumonia.[38, 46] In sarcoidosis, males tend to present with more advanced radiographic disease[26], however females more commonly demonstrate worse outcomes by other measures, including lower lung function,[18] higher rates of hospitalization,[28] higher disability utilization following diagnosis,[12] worse HRQoL,[47] and more depression[19] and anxiety.[48] Interestingly, males with IPF and sarcoidosis are more likely to be prescribed disease modifying agents than females.[15, 49]

Geographic Disparities

The prevalence and outcomes of patients with ILD and sarcoidosis vary by location.[50-52] US regional sarcoidosis mortality also intersects with race, demonstrating the highest age-adjusted death rates among Black females in the Midwest, and Black males in the Northeast. [53] Neighborhood-level disadvantage impacts the incidence of sarcoidosis,[54] and associates with higher mortality and lower odds of lung transplant for IPF.[55] Residents of rural areas with IPF present to ILD specialty care with higher oxygen utilization and lower pulmonary function compared to peers living in suburban areas.[56] Residents of rural areas with sarcoidosis are hospitalized more frequently, suggesting higher disease severity and/or access barriers.[8] Indeed, rural counties may lack hospital or pulmonary rehabilitation services.[57] Patients living farther from lung transplant centers are less likely to be listed for transplant in the United

States.[58] Interestingly, once listed for transplant, waitlist outcomes are better among patients who travel more than 360 miles to the center,[59] suggesting some favorable factors in extremely long-distance patients who have the resources to navigate the system.

Determinants of Disparities

While disparities are apparent, driving factors are incompletely elucidated. Access to care, stress, and environmental exposures are likely determinants of health disparities in both ILD and sarcoidosis. More investigation is needed to identify additional determinants.

Access to health care in the US has improved over the last decade, however gaps remain.[60] Limited access to pulmonary specialists associates with worse respiratory health outcomes.[61] In IPF, delay in access to a specialty center associates with increased mortality.[62] Individuals with sarcoidosis report access challenges,[11] however, more research is needed to understand the impact of limited access to care on sarcoidosis outcomes.

Patients with ILD and sarcoidosis report high stress levels.[63, 64] Low socioeconomic status,[65] racial discrimination,[66] gender discrimination,[67] and neighborhood factors[68] all associate with exacerbated stress. Higher stress levels associate with worse health outcomes including higher mortality,[69] higher disease burden,[70] and worse HRQoL.[71] Chronic stress upregulates pro-inflammatory gene transcription, [72] and links to glucocorticoid receptor resistance.[73] This results in increased pro-inflammatory cytokines including TNF-alpha which is thought to be part of the pathobiology of both ILD[74] and sarcoidosis.[75] More work is needed to understand the impact and biology of psychosocial and environmental stress on health outcomes and disparities in both ILD and sarcoidosis.

Socioeconomic and racial disparities persist in exposure to environmental toxins.[76] Disadvantaged neighborhoods suffer higher exposure to air pollution from industry, tobacco smoke, and automotive traffic, particularly in more densely populated areas.[77] Occupational exposures,[20] air pollution exposure,[20] and inhalation exposures[78] are frequently reported by patients with ILD and associate with lower lung function.[79] The role of tobacco smoke exposure in ILD cannot be understated. A study of idiopathic interstitial pneumonias found the odds ratio for ever smoking cigarettes to be 3.6 (1.3–9.8) in those who developed ILD,[80] suggesting the importance of tobacco exposure to disease onset. Poor air quality from motor vehicle exhaust, industrial, biomass, and indoor pollution has been associated with the production of reactive oxygen species, decreased telomere length, and release of inflammatory cytokines leading to complex interactions with patients' genetic predispositions and lung damage.[81]

Environmental factors such as occupation, the household environment, and lifestyle associate with the risk of developing sarcoidosis and respiratory symptom severity.[82] Interestingly, one study of Black American siblings found that indoor exposure to high humidity, water damage, or musty odors was associated with a higher odds of developing sarcoidosis.[83] Few studies detail the impact of environmental exposures on sarcoidosis progression.[42] Increased respiratory symptom severity has been associated with increased levels of air pollutants among individuals with fibrotic pulmonary sarcoidosis.[84]

Avenues to Address Disparities

In ILD, there have been few interventions to address disparities.[85] We are aware of no interventions conducted to address determinants of health disparities in sarcoidosis. However, there is a body of intervention research available to address disparities in other disease outcomes which may be leveraged and tested. Avenues to address the immense health

disparities in ILD and sarcoidosis include: 1. Health policy that promotes environmental justice and institutional and structural changes to address racism, 2. Research innovation and inclusion, and 3. Collaborative clinical care and patient-centered programs (Figure 3).

Policy level interventions to improve health disparities in ILD and sarcoidosis are an avenue to address disparities. Environmental policies that promote environmental justice are essential given the poor health outcomes in patients with ILD and sarcoidosis associated with environmental exposures. Confronting policies in the US that contribute to institutional and structural racism is also imperative to addressing health equity for ILD and sarcoidosis. Importantly, there remain challenges to implementation even after policy change. Despite the 1993 NIH requirement for inclusion and reporting of women and minorities in NIH-funded clinical trials, there has not been a significant increase in clinical trial reporting results by sex, race or ethnicity.[86] This suggests that while policy changes are important, successful implementation is necessary to impact equity.

We recognize the need to align the gender, race/ethnicity, and socioeconomic characteristics of clinical research participants with the population the research serves, and to improve diversity in care providers and investigators as well. Research statements by the American Thoracic Society[87] and National Institutes of Health[88] offer evidence-based solutions to improving recruitment and retention of participants from populations that remain underrepresented in research. Efforts to build a diverse pipeline of investigators and clinicians are underway.[89]

In addition to improving representation in clinical and translational research, methods of research including variables of interest must be carefully scrutinized and results interpreted with care. Self-identified race may correlate with geographic ancestry contributing to genomic variation,[90] but is primarily an intricate social construct. As a bundle of complex interacting relationships race warrants an equally complex analysis.[91] Advanced statistical support, which

stems from greater investment in research infrastructure and methods, is necessary to appropriately dissect the package of characteristics captured by racial identification.[92] Gender also warrants special attention as it encompasses the social constructs of masculinity and femininity and thus how patients experience the world and their interaction with medicine and research[36]. Gender should be considered apart from biological sex and also modeled as a complex variable. Fully understanding the biologic and social constructs of race, ethnicity, sex and gender in populations with ILD and sarcoidosis will allow for the development of actionable and clear interventions for improving disparities.

Research innovation in interventions utilizing home-based monitoring and telemedicine is one bright path forward. Global practice patterns of expert ILD clinicians surveyed in early 2021 found that 39% used some form of home monitoring. Those that utilized home monitoring rated their delivery of care as good as or even more effective than with face-to-face visits.[93] Comprehensive home monitoring added to usual care in newly treated patients with IPF does not improve overall HRQoL but significantly improves psychological well-being.[94] An area of particular enthusiasm with clinical and research design importance is home spirometry, where studies demonstrate good adherence, feasibility and reliability of home spirometry for individuals with ILD[94-96] and/or sarcoidosis.[97] Importantly in rare diseases such as ILD and sarcoidosis, home spirometry or other remotely and repetitively collected data can enhance precision of and power to detect effect, and thus diminish the patient sample size needed to detect a significant difference in measured outcome.[98] Further research is needed to understand how remote programs can be utilized to maintain and improve health.

Telemedicine boasts advantage to those who live distant from specialized care, or who lack time and resources to attend frequent in-person visits. However, equity in implementation of internet based-services and research can be stymied by gaps in technology access. One in four adults with low income are smartphone-only internet users, and a 35% gap exists between low

and middle-income household access to broadband internet.[99] Differences in internet access also exist between racial groups, educational levels, rural versus urban homes, and age groups.[99] Digital inclusion and equity is considered a super social determinant of health, as it directly impacts other social determinants of health, from jobs, to education, to medical care.[100] Nevertheless, early studies in telemedicine demonstrate promise not only for ease of patient interaction, but also for early detection of progressive disease.

Diagnostic complexity, delays and misdiagnosis plague patients with ILD[62] and sarcoidosis,[11] and post-diagnosis are followed by complexity of disease management.[6] Complex care models prioritizing comprehensive multidisciplinary approaches in ILD and sarcoidosis are needed.[101] Models including multidisciplinary discussion and specialized clinics report notable success, [102] and can be offered remotely to increase external access.[103] Dedicated ILD clinics not only improve survival but do so without increasing resource utilization.[102] To improve awareness of clinics and centers with expertise, patient-advocacy organizations such as the Pulmonary Fibrosis Foundation (PFF)[104] and the Foundation for Sarcoidosis Research (FSR)[105] have developed clinic and center of excellence designations; however, access to these specialized clinics remains a barrier. In the US as of August 2022, 16 states lack a PFF center and 20 states lack an FSR-recognized sarcoidosis clinic. Until additional centers and clinics are opened, interventions such as Project ECHO (Extension for Community Healthcare Outcomes), can connect providers with specialists through tele-mentoring to improve patient outcomes.[106] Innovation in complex clinical care and collaboration with patient advocacy organizations is imperative to ensure access to high quality care for all.

Self-management programs improve health outcomes,[107] however engagement in self-management is not uniform and associates with significant time investment for some patients.[108] Socioeconomic status,[109] race/ethnicity[109] and cultural beliefs and norms[110] influence individuals' engagement with self-management programs. Ensuring self-management programs utilize low

health literacy education, which improves outcomes,[111] is imperative. Designing self-management programs in ILD and sarcoidosis requires investigation of the complex factors that influence behaviors around self-management.

Conclusion

The roster of health disparities in ILD and sarcoidosis encompasses socioeconomic, race/ethnicity, sex/gender, and geographic differences. Flurries of recent publications document disparities, but we stand at the dawn of understanding and intervening upon the drivers. ILD and sarcoidosis sorely need energetic funding of cross-institutional and disciplined research, respiratory health policy making, and multi-faceted patient-centered care. Patient advocacy groups, researchers, clinicians and policy makers must all have seats at the table and work collaboratively in a team approach.

Figure 1: Treatment Access and Outcomes are Worse for Low-Income Individuals with Idiopathic Pulmonary Fibrosis (IPF)

Data from Swaminathan AC, Hellkamp AS, Neely ML, et al. Disparities in Lung Transplant among Patients with Idiopathic Pulmonary Fibrosis: An Analysis of the IPF-PRO Registry. Annals of the American Thoracic Society. Jun 2022;19(6):981-990; Sesé L, Caliez J, Annesi-Maesano I, et al. Low income and outcome in idiopathic pulmonary fibrosis: An association to uncover. Respir Med. Jul 2021;183:106415; and Dempsey TM, Payne S, Sangaralingham L, Yao X, Shah ND, Limper AH. Adoption of the Antifibrotic Medications Pirfenidone and Nintedanib for Patients with Idiopathic Pulmonary Fibrosis. Ann Am Thorac Soc. Jul 2021;18(7):1121-1128.

Figure 2: Black individuals are disproportionately affected by sarcoidosis

Data from Sharp M, Psoter KJ, Balasubramanian A, et al. Heterogeneity of Lung Function Phenotypes in Sarcoidosis: Role of Race and Sex Differences. Annals of the American Thoracic Society. Aug 4 2022; Judson MA, Boan AD, Lackland DT. The clinical course of sarcoidosis: presentation, diagnosis, and treatment in a large white and black cohort in the United States. Sarcoidosis, vasculitis, and diffuse lung diseases : official journal of WASOG. Oct 2012;29(2):119-27; Gerke AK, Yang M, Tang F, Cavanaugh JE, Polgreen PM. Increased hospitalizations among sarcoidosis patients from 1998 to 2008: a population-based cohort study. BMC pulmonary medicine. Jul 9 2012;12:19; and Mirsaeidi M, Machado RF, Schraufnagel D, Sweiss NJ, Baughman RP. Racial difference in sarcoidosis mortality in the United States. Chest. Feb 2015;147(2):438-449.

Figure 3: Avenues to Address Disparities

Clinics Care Points

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Treatment Access and Outcomes are Worse for Low-Income Individuals with Idiopathic Pulmonary Fibrosis (IPF)

IPF patients are **20% more likely** to receive lung transplantation for every $10,000 increase in income[21]

Patients pay an average of **$400 per month** for antifibrotic medications[15]

Low-income individuals with IPF are **50% more likely to experience early death**[20]

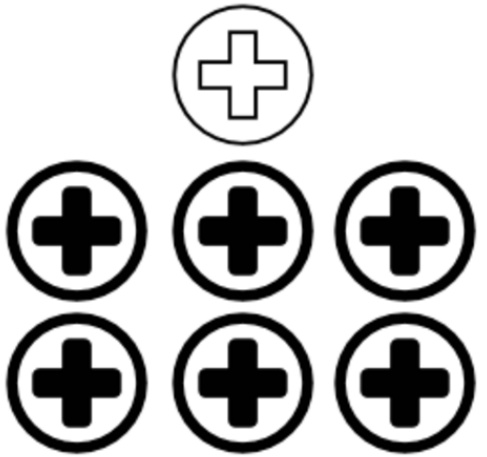
Data from Swaminathan et al. 2022, Annals ATS, Sesé et al, 2021, Resp Med, and Dempsey et al., 2021, Annals ATS.

Black individuals are disproportionately affected by sarcoidosis

Black individuals are more likely to have **multi-organ** disease[26] and **more severe** disease[7]

Data from Sharp et al. 2022, Annals ATS, Judson et al., 2012, JWASOG, Gerke et al., 2012, BMC Pulm Med, and Mirsaedi et al., 2015, Chest

Black individuals are **12 times** more likely to die from sarcoidosis[29]

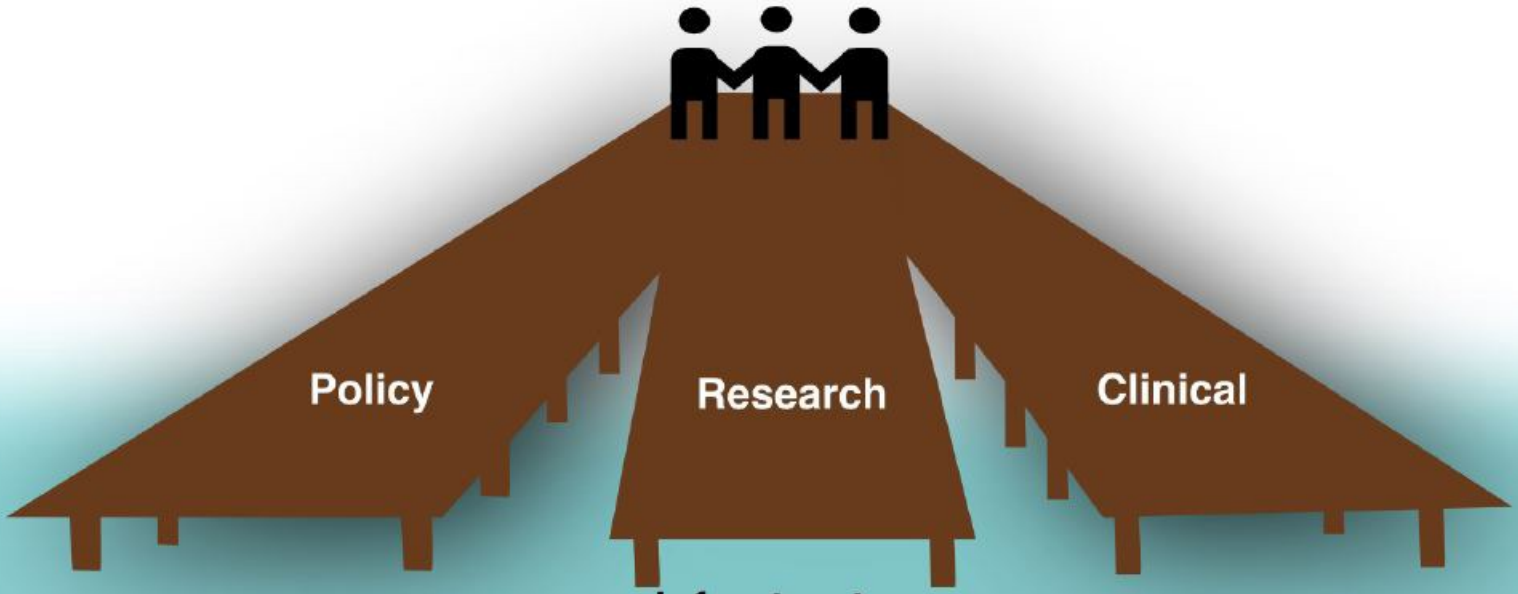


Black individuals are **6 times** more likely to be admitted to the hospital for sarcoidosis related complications[28]

White individuals with sarcoidosis

Black individuals with sarcoidosis

Interstitial Lung Disease & Sarcoidosis: BRIDGE TO EQUITY



Infrastructure
-Funding
-Industry Engagement
-Community Support and Engagement